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LESSON PLAN

For

AI Security for Autonomous AI Agents

TGS Ref No: TGS-2025060473

Conducted by

Tertiary Infotech Pte Ltd

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Version 2.0

Document Version Control Record

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	16 June 2025	Initial release — Core Principles and Ethical Challenges in Generative AI.	Dr Alfred Ang
2.0	17 August 2026	Course revised and retitled to AI Security for Autonomous AI Agents. Content rebuilt around AI security for generative AI and autonomous agents (OWASP LLM Top 10 2026, OWASP ASI Top 10 2026, NIST AI RMF, MITRE ATLAS, IMDA Model AI Governance for Agentic AI, PDPA/PDPC). Five real-world case-study activities added. Assessment restructured: Written Assessment (one question per K) and Case Study (one question per LO). Accredited TSC K/A statements unchanged.	Dr Alfred Ang

Table of Contents

TABLE OF CONTENTS

Right-click and choose “Update Field”, or press F9, to build the table of contents.

1. Course Overview

AI Security for Autonomous AI Agents is a 2 Days · 16 Hours WSQ course that equips learners to identify, analyse and mitigate the security risks introduced by generative AI systems and by autonomous AI agents. The course is delivered through real-world case studies drawn from 2026 incidents and is grounded in the current security and governance frameworks, including the OWASP Top 10 for LLM Applications (2026), the OWASP Top 10 for Agentic Applications (2026), the NIST AI Risk Management Framework, MITRE ATLAS, the IMDA Model AI Governance Framework for Agentic AI, and Singapore's PDPA together with the PDPC guidelines on personal data in generative AI.

The course covers AI security for BOTH generative AI (prompt injection, data leakage, poisoning, PDPA exposure) and autonomous agents (uncontrolled and destructive execution, tool misuse, agent vulnerabilities and cyber-attack chains).

Skills Framework Alignment

Element	Detail
TSC Title	Generative AI Principles and Applications
TSC Code	ICT-INT-0052-1.1
Proficiency Level	Level 1
Course Code	TGS-2025060473
Duration	2 Days · 16 Hours
Delivery	Instructor-led classroom with facilitated case-study activities

Note: the accredited knowledge and ability statements below are reproduced verbatim. They are taught and assessed as accredited; AI security is the delivery context through which each statement is evidenced.

Learning Outcomes

LO	Learning Outcome
LO1	Demonstrate generative AI concepts and applications relevant to customer service and hospitality management.
LO2	Apply prompt engineering techniques and analyse output variations to improve generative AI performance in service settings.
LO3	Identify ethical risks and analyse bias in AI-generated content used in customer engagement.

Knowledge Statements (assessed by the Written Assessment)

Code	Knowledge statement
K1	Importance of data quality, preprocessing, model pipeline and model training (e.g., impact of data bias from training data)
K2	Underlying principles, core concepts and theories governing generative AI
K3	Difference between generative and discriminative models

K4	Impact of prompt engineering on the model outputs of generative AI
K5	Generative AI model workings, including training data, algorithms, and outputs

Ability Statements (assessed by the Case Study)

Code	Ability statement
A1	Analyse limitations and potential biases in AI-generated content
A2	Identify the ethical implications and societal impact of AI-generated content
A3	Apply understanding of generative AI principles to use cases
A4	Demonstrate the use of generation AI in diverse applications (e.g., summarisation, inference, reasoning, transformation of content, augmentation of content)
A5	Analyse generative AI models' performance metrics and evaluate the influence of prompt variations

2. Daily Schedule

Each training day runs 9:30am – 6:30pm and delivers 8 instructional hours. A 1-hour lunch break is excluded from instructional time; short tea breaks are counted within the day.

Day 1 — Generative AI Security

Slides 1–30 of the trainer deck.

Time	Topic / Activity	Duration
9:30 – 9:45	Welcome, digital attendance (AM), introductions and ground rules	15 min
9:45 – 10:00	Course overview, learning outcomes and assessment briefing	15 min
10:00 – 10:45	LU1 T1: The AI security threat landscape — why generative AI breaks classical security assumptions (K2)	45 min
10:45 – 11:00	Tea break	15 min
11:00 – 11:45	LU1 T1 (cont.): The context window has no trust boundary; prompt injection as an architectural property (K2)	45 min
11:45 – 12:30	LU1 T2: Generative vs discriminative models — the guardrail classifier pattern and its limits (K3)	45 min
12:30 – 1:30	Lunch break	60 min
1:30 – 2:15	LU1 T3: Application modes as attack surfaces — summarisation, inference, reasoning, transformation, augmentation (A4)	45 min
2:15 – 3:00	Activity 1: Threat Modelling a Generative AI Concierge (K2, K3, A4)	45 min
3:00 – 3:45	LU2 T1: Data quality, preprocessing and the model pipeline — poisoning across training, RAG, memory and artifacts (K1)	45 min
3:45 – 4:00	Tea break	15 min
4:00 – 4:45	LU2 T2: Prompt injection in depth — direct, indirect and cross-modal; why prompt-based defences fail (K4)	45 min
4:45 – 5:45	Activity 2: Prompt Injection and the PDPA-Reportable Leak (K4, K1)	60 min
5:45 – 6:15	Activity 2 debrief — notification decision and prioritised controls	30 min
6:15 – 6:30	Day 1 recap, Q&A and PM digital attendance	15 min

Day 1 instructional total: 480 minutes (8 hours), excluding the 1-hour lunch break.

Day 2 — Autonomous Agent Security

Slides 31–64 of the trainer deck.

Time	Topic / Activity	Duration
9:30 – 9:40	Day 1 recap and AM digital attendance	10 min
9:40 – 10:25	LU2 T3: Security frameworks for generative AI and agents — OWASP LLM Top 10 2026, OWASP ASI Top 10 2026, NIST AI RMF, MITRE ATLAS, IMDA and the PDPA (A3)	45 min
10:25 – 11:00	LU2 T4: Measuring whether a guardrail works — attack success rate, refusal rate, false-positive rate across prompt variants (A5)	35 min
11:00 – 11:15	Tea break	15 min
11:15 – 12:15	Activity 3: Selecting a Security Framework for GenAI and Agents (A3, A5)	60 min
12:15 – 12:45	LU3 T1: Agent anatomy — model, loop, tools, memory, identity. Excessive agency, uncontrolled and destructive execution (K5)	30 min
12:45 – 1:45	Lunch break	60 min
1:45 – 2:00	LU3 T1 (cont.): Defence in depth and the four-layer control stack (K5)	15 min
2:00 – 3:00	Activity 4: Rogue Agent Post-Incident Review — the real 2026 incidents (K5)	60 min
3:00 – 3:30	LU3 T2: Ethical implications and societal impact — PDPA, PDPC GenAI guidelines and the IMDA Model AI Governance Framework for Agentic AI (A2)	30 min
3:30 – 3:45	Tea break	15 min
3:45 – 4:05	LU3 T3: Limitations and bias — misinformation as a security risk, aggregate accuracy and disparate impact (A1)	20 min
4:05 – 4:30	Activity 5: Agent Governance and the Deployment Gate — capstone (A1, A2)	25 min
4:30 – 4:40	Course synthesis, Q&A and briefing for assessment	10 min
4:40 – 5:40	Assessment 1: Written Assessment (SAQ) — 5 questions · 40 marks · 60 min	60 min
5:40 – 6:20	Assessment 2: Case Study — 3 questions · 70 marks · 40 min	40 min
6:20 – 6:30	Assessment digital attendance · TRAQOM survey · course close	10 min

Day 2 instructional total: 480 minutes (8 hours), excluding the 1-hour lunch break.

3. Activities

Every activity is a real-world case study. Each has its own folder under activities/ containing the scenario, the discussion questions, the trainer debrief and a printable PDF. Full step-by-step facilitation detail is in the Learner Guide.

#	Activity	Day	Duration	K/A assessed
1	Threat Modelling a Generative AI Concierge	1	45 min	K2, K3, A4
2	Prompt Injection and the PDPA-Reportable Leak	1	60 min	K4, K1
3	Selecting a Security Framework for GenAI and Agents	2	60 min	A3, A5
4	Rogue Agent Post-Incident Review	2	60 min	K5
5	Agent Governance and the Deployment Gate (capstone)	2	25 min	A1, A2

4. Tools and Resources

Resource	Purpose
Trainer slide deck (PPTX/PDF)	Facilitation, diagrams and case-study prompts
Learner Guide	Detailed step-by-step notes and activity walkthroughs
Activity packs (activities/)	Scenario, discussion questions and debrief per activity
LMS/TMS portal	lms-tms.tertiaryinfotech.com — course material and submission
Whiteboard / flip chart	Group threat modelling and kill-chain reconstruction
OWASP LLM & ASI Top 10 (2026)	Reference taxonomies used throughout
IMDA / PDPC publications	Singapore governance and data-protection reference

5. Assessment

Assessment is conducted at the end of Day 2. The briefing for assessment is delivered before the assessment begins.

Instrument	Covers	Detail
Written Assessment (SAQ)	K1 – K5	Written Assessment (SAQ) — 5 short-answer questions, one per K statement
Case Study	A1 – A5 via LO1 – LO3	Case Study — 3 questions, one per Learning Outcome, mapped to the A statements

Format	—	Open Book
Grading	—	Competent / Not Yet Competent
Re-assessment	—	Available for learners assessed Not Yet Competent

Funding eligibility requires a minimum 75% attendance recorded through SSG digital attendance, an assessment outcome of Competent, and completion of the TRAQOM survey.